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Comparing Housing Valuation Techniques and Stacked Generalization: Exploiting Explainable AI

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Abstract
This study investigates the use of diverse model selections in stacked generalization to improve the predictive accuracy of Automated Valuation Models (AVMs). The research focuses on three commonly used valuation techniques: the Comparable Sales Method (CSM), the linear Least Absolute Deviation (LAD), and the nonlinear XGBoost (XGB). A dataset consisting of 164,619 apartment transactions from Oslo between 2008 and 2022 is utilized for testing, with 25% of the data used for out-of-sample predictions. While the stacked model combining XGB, CSM, and LAD achieves the best performance with a Median Absolute Percentage Error (MdAPE) of 5.17%, the individual XGB model performs nearly as well, achieving an MdAPE of 5.24%. Analysis reveals that stacking provides marginal improvements and primarily relies on XGB predictions. However, the computational cost of stacking raises questions about its practicality. The research highlights the limitations and benefits of different housing valuation techniques and varying data sizes, offering practical insights to enhance the performance of future AI-AVMs. By utilizing explainable AI, this study contributes to a better understanding of how different models collaborate and diverge, revealing their competitive advantages in house price valuation.

Keywords Forecasting · Explainable AI · AI-AVM · Housing valuation · Comparable sales method

JEL Classification C55 · R21 · R31

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Introduction

> *"Valuation […] means the procedure and technique of estimating the value of* ***specific property*** *at a* ***stated time*** *and* ***place****."* (Bonbright, 1937, p. 10).

Traditionally, real estate valuation has been expensive, time-consuming, and required expertise. However, Automated Valuation Models (AVMs) and AI-AVMs have transformed the valuation of dwellings, making it accessible, instant, and free for anyone. While AVMs have clear practical implications, they also highlight the importance of understanding different real estate valuation techniques to enhance AVM predictability. This paper focuses on three common valuation techniques: the Comparable Sales Method (CSM), the linear Least Absolute Deviation (LAD), and the nonlinear XGBoost (XGB). Our aim is to investigate whether stacked generalization can enhance the predictive performance of AVMs in the housing market and to leverage explainable AI to better understand valuation techniques.

Stacked generalization works by dynamically weighting the input from base models based on their performances on a validation set (Wolpert, 1992). Each base model captures unique patterns or insights from the data, and the metamodel learns the optimal combination of these base model predictions to generate the final output. By leveraging the diversity among the base models, stacked generalization can potentially improve performance compared to using a single base model. This approach benefits stakeholders such as real estate professionals, investors, and homebuyers by providing a more robust and reliable prediction model. It also helps us gain deeper insights into the benefits of different valuation techniques.

Machine learning techniques have been used in a variety of topics in the real estate literature, such as the construction of house price indices (e.g., Calainho et al., 2022; Francke et al., 2023; Xu & Zhang, 2022), valuation of curb appeal and architectural styles (e.g., Johnson et al., 2020; Lindenthal & Johnson, 2021; Wan & Lindenthal, 2023), and valuation of commercial real estate (e.g., Deppner et al., 2023). However, most studies have focused on integrating various machine learning methods to enhance predictability and to compare traditional approaches with machine learning methods in residential property valuation (e.g., Bogin & Shui, 2020; Simlai, 2021; Hjort et al., 2022; Ho et al., 2021; Mayer et al., 2019; Pace & Hayunga, 2020; Pérez-Rave et al., 2019; Sing et al., 2021; Yacim & Boshoff, 2018; Yoshida et al., 2022). Some studies have investigated the use of stacked generalization to improve prediction performance in real estate valuation (Birkeland et al., 2021; Fourkiotis & Tsadiras, 2023; Graczyk et al., 2010; Truong et al., 2020). These studies have demonstrated that models using stacked generalization not only enhance the stability of predictions, but also outperform other models in terms of predictive accuracy. Our study aims to extend the literature by investigating the implementation of stacked generalization with methodically diverse modeling techniques. Specifically, we integrate the CSM, a traditional appraisal method not commonly used in this context, along with the LAD as a linear model and the XGB as a non-linear model. The selection of diverse models ensures a wide range of approaches is considered and leverages the diversity among base models. By incorporating the CSM into the stacked generalization

framework, we explore a new direction in real estate valuation, shedding light on the potential benefits and challenges associated with this procedure and different valuation techniques.

We utilize a dataset comprising 164,619 apartment transactions in Oslo, covering the period from 2008 to 2022. Our models are trained on 75% of the data and tested on the remaining 25%. To address our research question, we develop AVMs for the three individual models and feed their predictions into an XGB meta-model as part of the stacked generalization process. This enables the creation of three additional stacked models: (XGB + CSM), (XGB + LAD), and (XGB + CSM + LAD). We compare these six models using various evaluation metrics and gain further insights by examining feature importance and the discrepancy between predicted and actual values. Additionally, we conduct tests to estimate the models' out-of-sample performance across smaller training sets representing different scenarios.

Our findings suggest that implementing stacked generalization with diverse models leads to only a marginal improvement in prediction performance. Notably, the XGB model performs robustly when used alone on the extensive housing market sample from Oslo. Our analysis also reveals that the stacked models primarily rely on the XGB base model for accuracy. However, the practicality of stacked generalization is a concern due to its computational expense. Nonetheless, integrating XGB with the CSM and LAD through stacked generalization yields a minor yet significant improvement across all performance metrics, reducing the Median Absolute Percentage Error (MdAPE) from 5.24 to 5.17%. When evaluating out-of-time test samples, the overall performance drops as expected. However, the models' relative performance remains largely consistent, and our main conclusion holds despite the somewhat less distinct results.

When examining models trained on smaller samples, we observe that the linear LAD model improves its MdAPE performance until reaching a training sample size of approximately 15,000 observations. Most of the prediction improvement for XGB has already occurred around 20,000 observations, but it continues to show gradual improvement with larger training samples. Not surprisingly, the accuracy of the CSM heavily relies on a substantial amount of data. The CSM's performance surpasses that of the linear model when the training sample reaches around 30,000 observations. This aligns with our expectations as the linear model fails to fully utilize the available information, particularly regarding location and sales date. The CSM performs poorly with small training samples because it relies on information from dwellings that significantly deviate from the property being valued. As the training sample size increases, the model can base its valuation on more similar properties, like XGB, effectively leveraging the available information for each individual observation.

The remainder of this study is structured as follows: Sect. 2 establishes the theoretical foundation through a literature review. Section 3 presents the data. Section 4 illustrates the development of our AVMs, including stacked generalization, individual models and the SHapley Additive exPlanations (SHAP) framework for feature importance. Section 5 evaluates and discusses the models' performance under various conditions. Finally, Sect. 6 concludes.

Literature Review

Stacked Generalization

Stacked generalization is an ensemble learning technique that combines predictions from multiple base models to create a final model with improved accuracy and robustness. It was first introduced by Wolpert (1992) as a solution to the limitations of single-model approaches in classification tasks. Breiman (1996) further refined the technique for regression tasks and discovered that using diverse models led to better results. Smyth and Wolpert (1997) extended the application of stacked generalization to unsupervised learning tasks, expanding its potential beyond supervised learning. Their work showcased the versatility of stacked generalization and contributed to advancements in the field of machine learning, opening up new possibilities for its utilization in various unsupervised learning tasks.

In real estate valuation, stacking has been implemented in several studies. Graczyk et al. (2010) explored the effectiveness of six algorithms, including neural networks, decision trees, linear regression, and support vector machines, as base models in ensemble learning for real estate appraisal. They compared three meta-learners: Additive Regression, Bagging, and Stacking, in terms of prediction accuracy. The results showed that bagging was the most stable technique, while stacking achieved the lowest prediction errors in most cases. Truong et al. (2020) focused on predicting house prices in Beijing using three machine learning models: Random Forest (RF), XGB, and LightGBM. They then compared the performance of these individual models against a stacked model using stacked generalization. The findings indicated that RF outperformed the stacked model on the training set and was the best-performing model among the three. However, it was observed that RF tended to overfit, while the stacked model demonstrated superior performance on out-of-sample data, suggesting its potential for improved generalization.

Birkeland et al. (2021) proposed an Automated Valuation Model (AVM) for the Oslo residential real estate market using a stacked ensemble approach. The ensemble combined base models like Bagging Predictor (BP), Random Forest (RF), Extra Trees (ET), and XGB, with another XGB model serving as the meta-model. The stacked ensemble outperformed the individual models when evaluated on out-of-sample data, with low MdAPE and a high percentage of predictions within 20% of the true values. However, the authors acknowledged that the stacked generalization approach was computationally expensive and might present challenges in interpreting the output due to potential hidden factors influencing the model's predictions. To gain further insights, incorporating Linear Regression (Kansara et al., 2018) and ElasticNet Regression (Nnadozie et al., 2022) alongside a subset of the aforementioned base models can be beneficial. XGB emerged as the most effective individual model in both studies, with the stacked model showcasing superior performance. Furthermore, Fourkiotis and Tsadiras (2023) found that ensemble methods are valuable because of their ability to harness the advantages of multiple individual algorithms.

Comparable Sales Method and its Variations

CSM involves analyzing recent sales of comparable properties located in the same area as the target property, and adjusting the sales prices based on differences in various factors such as size, location, age, and other relevant features. The traditional CSM model is effectively a constrained spatial autoregression (Pace & Gilley, 1998). In its simplest form, commonly referred to as an expert algorithm, it imitates the appraisal process undertaken by professional appraisers. The algorithm finds comparables that match the target property in its characteristics and computes the average value of their transaction price to estimate a prediction for the target property. Examples of using such an approach in recent literature include Larraz et al. (2021), Stang et al. (2022) and Trawiński et al. (2017), albeit with results that were surpassed by more complex models. However, Stang et al. (2022) showed that their Expert Function (EXF) was better suited to estimate the price in some of the districts in Germany, especially those with fewer observations.

An alternative approach is to use a K-Nearest Neighbors (KNN) algorithm to calculate distances between the sample and target property to find a number (K) of similar properties. Antipov and Pokryshevskaya (2012) and Valier (2020) obtained more promising results using this approach compared to the simpler expert algorithm, although the tree-based machine learning algorithms still yielded better results. Building upon this approach, Oust et al. (2020) proposed a variation by incorporating a K-means algorithm as a function of longitude, latitude and price per square meter on the training set to generate more homogeneous artificial districts. Subsequently, KNN was applied to the test set to classify dwellings based on the newly constructed districts, employing the Haversine formula to measure distance. This resulted in hedonic regressions that were spatially enhanced, effectively serving as a CSM. Birkeland et al. (2021) utilized a similar method with the Haversine formula, known as the comparable pricing method, to train ensemble learning submodels of an AVM for estimating the value of different types of dwellings. The method involves selecting geographically nearest transactions of the same type of dwelling as the target property, with a ranking variable based on proximity, enabling the ensemble methods to recognize recent sales of similar properties located in the near vicinity.

Hedonic Regression Models

One of the earliest and most widely used AVMs is the Hedonic Pricing Model (HPM) proposed by Rosen (1974). This model is often used as a benchmark for testing other models. Recent studies, such as the one conducted by Doumpos et al. (2021), have shown that even simple linear models can be effective tools for building accurate AVM systems, as long as they account for spatial effects. The study found that their locally weighted linear regression models, including Ordinary Least Squares (OLS), Least Absolute Shrinkage and Selection Operator (LASSO), and LAD, performed well in capturing spatial price variation compared to global and unweighted local approaches. These models were also found to be easy to implement in a large-scale context and required only a small data sample for model fitting. LAD was first introduced by Koenker and Bassett Jr (1978) as an alternative to other more commonly used linear

regression models. Yoo (2001) found that the LAD model generates more robust and accurate estimates of hedonic regression models compared to the conventional OLS method. These findings demonstrate the advantages of using the LAD method for HPM analysis and its potential to provide more reasonable results and increased robustness for such models.

Boosting Techniques

Boosting is an ensemble technique first introduced by Schapire (1990). It involves training decision trees sequentially, where each subsequent tree is fitted to improve the errors made by the preceding trees. Later, this technique was developed further and the AdaBoost method was introduced by Freund and Schapire (1997), which was widely adopted in various machine learning applications. Gradient Boosting Machine (GBM) was developed by Friedman (2001) as a generalization of AdaBoost to achieve higher accuracy, handle complex interactions between variables, and optimize any differentiable loss function. However, this algorithm can be computationally expensive and can also lead to overfitting. To address these issues, Chen and Guestrin (2016) implemented the eXtreme Gradient Boosting (XGB) algorithm. The model is a powerful and computationally efficient implementation of the gradient boosting model. It is designed to address overfitting issues by imposing regularization on the model structure, which limits its complexity and improves generalization for out-of-sample data.

The XGB model has since been popular in AVMs and is well documented in previous literature. Kumkar et al. (2018) conducted a comparative analysis of four tree-based ensemble methods, including XGB, to evaluate their efficiency in the appraisal of property values. The results demonstrate that the XGB model outperforms the other three models in terms of accuracy and predictive power. Stang et al. (2022) found that due to its flexibility the XGB function achieved the highest overall accuracy compared to the OLS regression, Generalized Additive Model (GAM), and their EXF in their AVM for real estate properties. Hjort et al. (2022) used a large dataset containing 126,719 observations where their variant of the XGB model achieved the best results on 4 out of 5 of the performance metrics compared to other models, where only their RF model performed better on median error percentage. Although XGB has been shown to be a superior model in some contexts, it has been reported to be outperformed by at least one other model in various comparisons (Birkeland et al., 2021; Law et al., 2019; Sangani et al., 2017; Wang et al., 2020).

Data

Our dataset comprises 199,370 residential transactions in Oslo, spanning the period between 2008 and 2022, provided by the Norwegian PropTech company Solgt.no. A significant majority of dwellings are identified as apartments, and by concentrating our analysis solely on these, we gain the advantage of working with a more homogeneous market, which is highly beneficial in the development of an AVM. The dataset contains missing values in certain variables and the possibility

of typographical errors in the source ads, requiring thorough attention during the cleaning process. Consequently, the focus is on addressing these issues, with careful consideration given to minimizing data loss when removing missing values. The approach is partially influenced by Helgaker et al. (2022), but some modifications are made to the process to retain a greater quantity of data. A summary of the full cleaning process is provided in Table 1.

As the dependent variable in our analysis, we use the natural log of the price per square meter of living area. Following the approach of Oust et al. (2020), we include both the actual transaction price and the common debt in the price to accurately capture the total cost associated with acquiring a property, thereby providing a more comprehensive representation of the property's market value.[1] While city districts are used to capture locational effects in the LAD model (see Fig. 1), we use latitude and longitude coordinates for the XGB. To calculate the distances in the CSM, we transform the coordinates from degrees to radians.

There is a considerable variation in prices across city districts. Figure 1 depicts the distribution of the price per square meter based on latitude and longitude coordinates, revealing a discernible trend: prices increase as the proximity to the city center intensifies. The five districts Gamle Oslo, Grünerløkka, Sagene, St. Hanshaugen and Frogner represent the inner city, where apartments are generally smaller due to their central location. This leads to a relatively higher price per square meter compared to

Table 1 Data pre-processing steps of the original dataset

Data pre-processing	Observations
Original data	199,370
Apartment	174,815
Detached house	10,331
Row house	7,898
Semi-detached house	6,066
Other	260
Keep only apartments	174,815
Keep observations with sales price more than 500,000	174,598
Keep observations with living area larger than 9 m^2 and less than 300 m^2	174,484
Keep observations with build year newer than 1600	174,198
Keep observations with less than 10 bedrooms	165,077
Remove observations with faulty coordinates	164,674
Keep observations with floor less than 21	164,671
Keep observations with less than 4 bathrooms	164,654
Remove observations within the district "Marka"	164,652
Remove observations from 2007	164,636
Remove observations with negative age	164,619

The final dataset consists of 164,619 observations.

[1] Common debt, or shared debt, refers to the portion of a loan or mortgage collectively held by all members of a housing cooperative or co-ownership. It finances common property expenses, such as building maintenance and renovations. The market value includes both the transaction price and the common debt, with the transaction price adjusting based on the level of common debt.

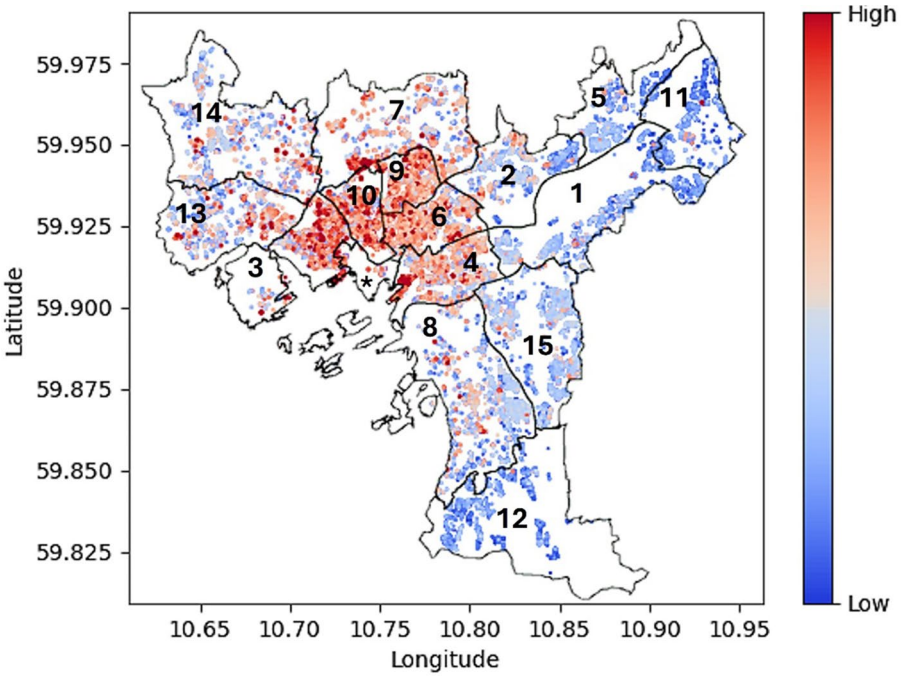

Fig. 1 Districts of Oslo and price per m2 by location. Note: Axes refer to latitude and longitude, while colors refer to relative prices. A small number of observations from the district "Sentrum", which lacks its own administration, were reallocated to the district "St. Hanshaugen". (1) Alna, (2) Bjerke, (3) Frogner, (4) Gamle Oslo, (5) Grorud, (6) Grünerløkka, (7) Nordre Aker, (8) Nordstrand, (9) Sagene, (10) St. Hanshaugen, 11) Stovner, 12) Søndre Nordstrand, 13) Ullern, 14) Vestre Aker, 15) Østensjø, *) Sentrum

other areas in Oslo. Additionally, the figure demonstrates a modest increase in prices for the western regions of Oslo, which can be attributed to the relatively greater affluence compared to the eastern regions.

To facilitate the implementation of the linear LAD model, we partition the sales date into its year and quarter components. For the CSM and XGB models, we construct a new variable referred to as *Days elapsed*, which records the number of days that have elapsed since the first transaction in the dataset. A collection of binary variables is used to provide insights into the apartment features. It is important to note that their inclusion in an ad is optional, and different real estate agents may have varying perspectives on which facilities to highlight as relevant. Therefore, a degree of uncertainty is associated with the facility variables, and it is plausible that some ads may lack certain facilities despite the underlying apartment possessing the attribute. However, after assessing the quality and relevance of each variable, they exhibit a distribution that appears reasonable. Additionally, we generate two new binary variables: one indicating whether the apartment is situated on the fifth floor or higher and lacks an elevator, and another indicating whether the apartment is located in any of the five inner city districts and situated on the first floor. Apartments

located on the first floor in the inner city are more likely to experience a negative price impact due to increased pedestrian flow and limited privacy.

Renovation object is a binary variable indicating whether certain keywords in the ad title suggest that the apartment requires renovation.[2] Although there may be some uncertainty associated with this variable, it is common to include this type of information in the ad title to attract certain types of homebuyers. A total of 10,938 transactions (6.64%) are categorized as renovation objects, which likely capture most apartments in need of renovation. Summary statistics are reported in Table 2.

Table 2 Summary statistics

Panel A: Numerical variables	Mean		SD	Min	Max	
Price per m^2 (NOK 1000)	65.17		23.29	12.50	278.33	
Size (m^2)	66.31		25.04	10.00	299.00	
Age (years)	59.46		38.70	0.00	372.00	
Bedrooms	1.78		0.76	1	9	
Bathrooms	1.05		0.21	1	3	
Latitude (degrees)	59.92		0.03	59.82	59.98	
Longitude (degrees)	10.78		0.06	10.63	10.95	
Latitude (radians)	1.05		0.00	1.04	1.05	
Longitude (radians)	0.19		0.00	0.19	0.19	
Days elapsed (1000)	3.49		1.38	0.00	5.68	
Panel B: Binary variables	%	N	Binary variables (cont.)		%	N
Balcony	68.00	111,941	Østensjø		6.43	10,580
Garage	36.34	59,822	Sales year			
Quiet	59.79	98,426	2008		1.56	2,563
View	35.00	57,614	2009		2.30	3,790
High floor and no elevator	5.05	8,321	2010		2.89	4,756
First floor and inner city	8.18	13,460	2011		4.81	7,919
Renovation object	6.64	10,938	2012		6.72	11,067
District			2013		7.18	11,812
Alna	5.88	9,677	2014		7.02	11,559
Bjerke	4.37	7,185	2015		7.84	12,906
Frogner	12.56	20,670	2016		7.19	11,837
Gamle Oslo	11.96	19,688	2017		7.49	12,333
Grorud	2.94	4,836	2018		8.36	13,763
Grünerløkka	13.36	21,985	2019		9.27	15,257
Nordre Aker	4.86	7,997	2020		9.53	15,689
Nordstrand	4.78	7,867	2021		10.16	16,731
Sagene	11.54	19,002	2022		7.67	12,637
St. Hanshaugen	8.77	14,443	Sales quarter			
Stovner	2.58	4,257	Q1		22.93	37,752
Søndre Nordstrand	2.47	4,066	Q2		30.44	50,116
Ullern	3.83	6,304	Q3		25.14	41,384
Vestre Aker	3.68	6,062	Q4		21.48	35,367

The variable *Days elapsed* counts the days that have elapsed since the first transaction in the sample.

[2] The variable is based on word recognition for the Norwegian equivalents of "remodeling" and "renovation" in the ad title.

Methodology

To prepare for the construction of predictive models, the dataset is divided into a training set and a test set using random allocation, with a 75/25 split. This allocation ensures that the models have access to a substantial amount of data for training, while also providing a sufficiently large sample size for evaluating their performance. The training and test sets comprise 123,464 and 41,155 observations, respectively. By utilizing different models, we aim to capture a wide range of features and relationships within the data, and to assess the relative strengths and weaknesses in each of the models.

The Base Models

CSM Model

The KNN algorithm serves as the fundamental methodology in our CSM model. We employ KNN for its scalability, allowing us to extract valuable information from the full sample, in contrast to other CSM approaches that typically involve filtering and constraining the study to a smaller sample. This approach involves identifying a set of K neighboring apartments that are most similar to the apartment in question and using them to estimate the target value, assuming that properties with similar features are likely to have similar values. The traditional CSM procedure involves making adjustments to the features of nearby sales to calculate the target value, much like how appraisers assess property values in real life. However, in this study's approach to CSM, the required adjustments are made through the selection process of the K nearest neighbors. Therefore, our CSM procedure approximates the traditional approach.

The KNN algorithm calculates the multidimensional distance between each property using the Euclidean distance formula, as defined in Eq. (1).

$$d\left(x, y\right) = \sqrt{\sum\nolimits_{i=1}^{n} w_i \times \left(x_i - y_i\right)^2} \tag{1}$$

The data points to be compared are represented by x and y, the feature values of the data points along each of the n dimensions are denoted by x_i and y_i, and the weights given to each dimension are denoted by w_i. The Euclidean distance metric is particularly useful in situations where the calculation must consider multiple features beyond just the spatial location. In our CSM model, we use 13 dimensions. In addition to the temporal (*Days elapsed*) and spatial dimensions (*Latitude* and *Longitude* in radians), these include *Size (ln)*, *Age (ln)*, *Bedrooms*, *Bathrooms* and a set of seven binary feature variables: *Balcony*, *Garage*, *Quiet*, *View*, *High floor and no elevator*, *First floor and inner city*, and *Renovation object*.

A limitation of the distance calculation described above is its sensitivity to scale and inability to prioritize features based on their importance. To address this, we first standardize the dataset to eliminate the effects of scaling. Additionally, to ensure that more important variables have a greater influence on the distance calculation,

we apply weights to the standardized features. These weights are derived from the coefficients of a simple OLS model fitted to the standardized training sample.[3] Features with larger coefficients are assigned higher weights, effectively amplifying their impact on the distance metric.

After calculating the distances, the KNN algorithm identifies the K nearest neighbors. To determine the optimal value for K, we employ grid search, a hyperparameter tuning technique, testing values ranging from 1 to 20 and utilizing mean absolute error (MAE) as the performance evaluation metric. Our experiments reveal that K = 10 is the optimal value, resulting in the lowest MAE. The 10 nearest neighbors are then used to estimate the natural logarithm of price per square meter by obtaining the weighted average of their values. This weighting is done with regards to the multidimensional distance, so that more similar neighbors are weighted higher.

LAD Model

We use the linear LAD method, which is equivalent to a median regression, instead of the more traditionally used OLS, due to its robustness in the face of outliers within the dataset, as noted by Yoo (2001). One potential issue with OLS estimations is the powerful influence of extreme values, as the loss function minimizes squared prediction errors. The LAD model mitigates this issue by minimizing absolute prediction errors, assigning relatively less weight to outlying observations. Following the standard hedonic model widely used in the real estate literature, the LAD model estimates the marginal impact of a set of K independent variables, X_k, on the natural logarithm of square meter price, $\ln(P_i)$, of an apartment, i, as presented in Eq. (2).

$$\begin{aligned}\ln(P_i) = \beta_0 &+ \sum_k \beta_k X_{k,i} + \sum_t \sum_q \gamma_{t,q} \left(\text{Year}_t \times \text{Quarter}_q\right) \\ &+ \sum_d \delta_d \text{District}_d + \sum_k \sum_d \eta_{kd} \left(X_{k,i} \times \text{District}_d\right) \\ &+ \sum_k \sum_t \psi_{kt} \left(X_{k,i} \times \text{Year}_t\right) + \epsilon_i\end{aligned} \tag{2}$$

The predictors, X_k, include S*ize (ln)*, *Age (ln)*, *Bedrooms*, *Bathrooms*, *Balcony*, *Garage*, *Quiet*, *View*, *High floor and no elevator*, *First floor and inner city*, and *Renovation object*.

To capture temporal and spatial variations, the model includes year-quarter fixed effects, $\gamma_{t,q}\left(\text{Year}_t \times \text{Quarter}_q\right)$, and district fixed effects, $\delta_d \text{District}_d$. Given that the sample period spans 15 years and covers 15 districts, it is plausible that price dynamics vary over time or across different districts. To address this, we interact the predictor variables X_k with both the district and year. These interactions enable the

[3] The weights given to each dimension, i.e., the coefficients of the OLS fitted to the standardized training sample, are the following, in ranked order: *Sales time* (0.3257), *Size* (0.2707), *Latitude* (0.2415), *Longitude* (0.2415), *Bathrooms* (0.0432), *Renovation object* (0.0402), *Bedrooms* (0.0363), *Age* (0.0276), *First floor inner city* (0.0263), *Garage* (0.0103), *High floor no elevator* (0.0070), *Quiet* (0.0064), *View* (0.0058), and *Balcony* (0.0029). The weights assigned to latitude and longitude are obtained by mean encoding the district variable, transforming it into a location variable, and then using it in the OLS regression after standardization.

effects of the predictors to vary spatially and temporally, providing greater flexibility in modeling localized and time-sensitive price dynamics. For the full in-sample estimation, please refer to Table 5 in the Appendix.

XGB Model

XGB operates by incorporating decision trees into a model iteratively, with each subsequent tree aiming to rectify errors made by previous trees. During training, the XGB model optimizes a loss function that measures the disparity between predicted and actual values (Chen & Guestrin, 2016). It uses gradient descent to update the weights of the decision trees and minimize the loss function by adjusting parameter values. To address overfitting, XGB employs regularization techniques by adding penalty terms to the objective function during training. These techniques discourage the model from fitting noise in the dataset and penalize model complexity. For further technical details, we follow the approach in Chen and Guestrin (2016).

To minimize MAE, which is chosen as the loss term for the objective function, and optimize performance, we utilize grid search and 5-fold cross-validation. Through grid search, we systematically train and evaluate the XGB model using various hyperparameter values within a predefined grid. This allows us to explore multiple combinations of hyperparameters and identify the best values for each parameter. By leveraging the 5-fold cross-validation, we ensure robustness and reliable estimation of model performance. The input variables in the XGB model are *Days elapsed*, *Latitude*, *Longitude*, S*ize (ln)*, *Age (ln)*, *Bedrooms*, *Bathrooms*, *Balcony*, *Garage*, *Quiet*, *View*, *High floor and no elevator*, *First floor and inner city*, and *Renovation object*.

Stacked Generalization

Stacked generalization exploits the strengths of the base models by combining them in a meta-model, which is trained on a new dataset consisting of predictions from the individual base models along with the XGB base model input variables. We use another XGB model as the meta-model based on its capability to handle a large number of features and capture non-linear relationships between input features and the target variable. It has also been proven well suited for this purpose in previous literature (Birkeland et al., 2021). The stacking process is visualized in Fig. 2.

To prevent overfitting in the stacked ensemble, we perform a 5-fold cross-validation for each base model by dividing the original training set into training folds and a validation fold. This allows us to train and evaluate the ensemble multiple times on different subsets of the data, ensuring that the models are not overfitting to a particular subset. Once the meta-model is trained, it is used to make predictions on a holdout test sample. We analyze three different meta-models from combinations of the base models: XGB + CSM, XGB + LAD, and XGB + CSM + LAD.

The three individual models and the three stacked models are evaluated by R-squared (R^2), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Median Absolute Percentage Error (MdAPE), and Percentage Prediction Error (PPE). To evaluate feature importance, we utilize the SHapley Additive exPlanations (SHAP) framework (Shapley, 1953; Strumbelj & Kononenko, 2010),

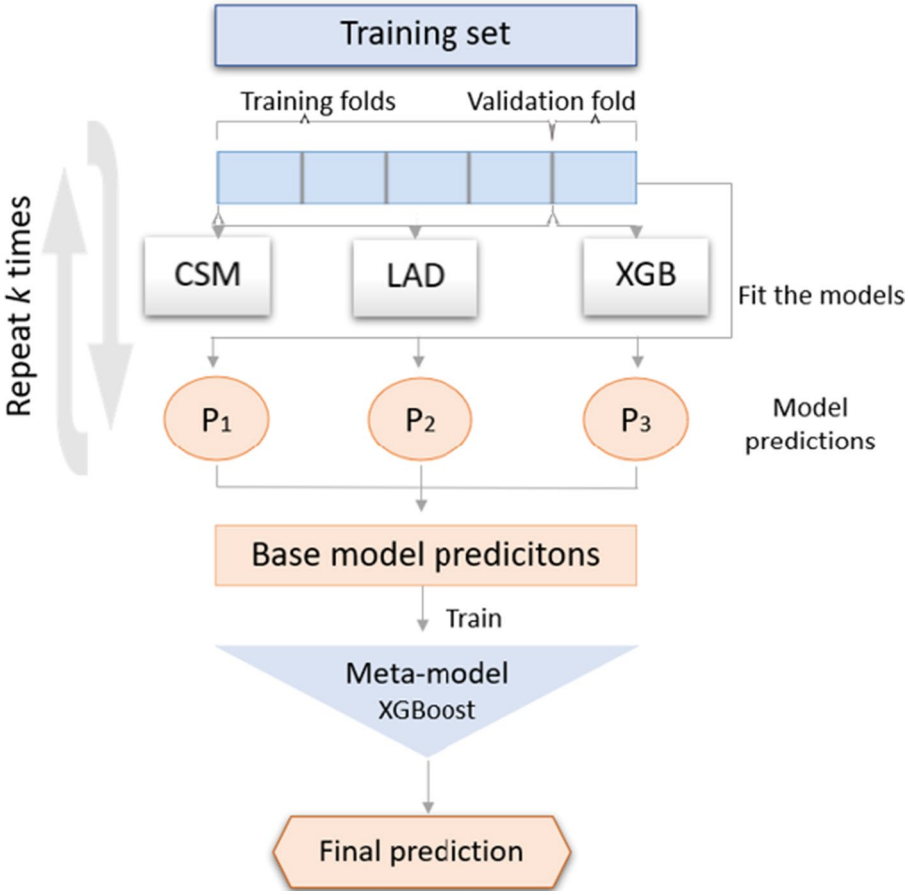

Fig. 2 The process of stacked generalization. Note: We use a 75/25 split for training and test samples

following Lundberg and Lee (2017). SHAP values assign an importance score to each feature by measuring its contribution to a specific model prediction. Based on cooperative game theory, SHAP values uniquely satisfy desirable properties such as local accuracy, consistency, and missingness, making them a unified measure of feature importance across various models (Lundberg & Lee, 2017). Feature importance plots provide a global view of how features influence model predictions by ranking them based on their average SHAP values across all predictions. In contrast, beeswarm plots display the distribution of SHAP values for each feature across all predictions, with each dot representing a single prediction. Beeswarm plots show not only the magnitude of feature importance but also the variability and direction of influence, with color indicating the feature value. However, obtaining SHAP values for non-linear or non-tree-based models, such as our CSM with a KNN algorithm, presents computational challenges. To address this, we employ randomly drawn background and test samples consisting of 300 and 100 observations, respectively, to estimate feature importances. Comparisons with the entire dataset for both LAD and

XGB models consistently yield similar variable rankings. Therefore, we consider our reduced random sample representative of the entire dataset, allowing for comparison across all models.

An alternative to the stacked generalization approach is Bayesian Model Averaging (BMA), a probabilistic framework that addresses model uncertainty by averaging over a set of candidate models, with each model weighted by its posterior probability (Hoeting et al., 1999). While BMA offers theoretical advantages, such as comprehensive uncertainty quantification, it has practical limitations, including computational intensity and sensitivity to prior assumptions, particularly in large model spaces (Hoeting et al., 1999; Steel, 2020). Furthermore, studies have shown that stacked generalization can outperform BMA when the true data-generating model is not included in the candidate set (Clarke, 2003; Yao et al., 2018).

However, by choosing stacked generalization, we forgo BMA's ability to provide a full posterior distribution, which is a powerful tool for uncertainty quantification (Steel, 2020). To address this limitation, we compute the Automated Valuation Model Uncertainty (AVMU) alongside standard performance metrics. Following the methodology of Pollestad et al. (2024), AVMU is defined as the average standard deviation of the estimated 90% prediction intervals for each model. This metric complements traditional performance metrics by providing both insights into model accuracy and a quantification of prediction uncertainty.

Importantly, these uncertainty estimates are generated ex-ante, without reference to actual sales prices in the test set. Initial heuristic uncertainty measures are derived using the direct loss estimation approach outlined in Pollestad et al. (2024) and are subsequently calibrated to match the standard deviations of error distributions using a conformal prediction framework. This approach enables the construction of prediction intervals for each AVM, including complex "black-box" models, with the resulting averages forming the AVMU metric. While AVMU provides an average measure for interpreting model uncertainty, the individual prediction intervals vary across homes, reflecting differences in uncertainty for specific predictions.

Results and Discussion

Model Performance in Out-of-Sample Prediction

Table 3 shows the performance of the individual and stacked models. Among the individual models, XGB performs best on all performance metrics, followed by CSM and LAD. When introducing stacked generalization, we observe that the combination of XGB and CSM performs only marginally better than the individual XGB, while the combination of XGB and LAD actually performs worse. The combination of all three base models seems to be the best overall, outperforming the other models on all metrics, except for a tie on PPE(50). However, the differences between the individual XGB and the three stacking combinations are only marginal. This indicates that the LAD and CSM models fail to bring any new valuable information that is not already covered by the XGB.

Table 3 Comparison of model performance

Models	R^2	RMSE	MAPE	MdAPE	PPE(10)	PPE(20)	PPE(50)	AVMU
CSM	0.9038	0.1124	8.36	6.51	69.15%	92.95%	99.78%	10.46%
LAD	0.8780	0.1266	9.25	7.22	64.18%	90.60%	99.74%	11.41%
XGB	0.9393	0.0893	6.69	5.24	78.46%	96.63%	**99.91%**	8.36%
(XGB + CSM)	0.9399	0.0889	6.66	5.21	78.69%	96.69%	**99.91%**	8.33%
(XGB + LAD)	0.9388	0.0897	6.71	5.25	78.32%	96.57%	**99.91%**	8.40%
(XGB + CSM + LAD)	**0.9403**	**0.0886**	**6.63**	**5.17**	**78.90%**	**96.73%**	**99.91%**	**8.31%**

R^2 and RMSE are calculated using the log-transformed observed and predicted values, while MAPE, MdAPE, PPEs, and AVMU are calculated after reverting values to their original scale (Price per m^2). Highlighted numbers indicate best performance.

To provide a more illustrative representation, Fig. 3 displays the distribution of relative prediction errors. The results indicate that XGB outperforms CSM and LAD, exhibiting a narrower error distribution and a higher proportion of errors within the dashed lines (+/- 20%), implying a superior model. The three stacked models appear nearly identical, exhibiting a distribution pattern similar to that of XGB alone, suggesting that the stacked models are predominantly reliant on the XGB algorithm. Overall, we observe that the XGB model performs exceptionally well on its own, raising the question of whether stacked generalization is worth the added complexity and resources required for training and prediction.

Figure 4 portrays the SHAP values derived from all models, offering a comprehensive overview of each variable across the models. As the models use distinct features to represent time, location, and size, we have aggregated the SHAP values to facilitate comparison.

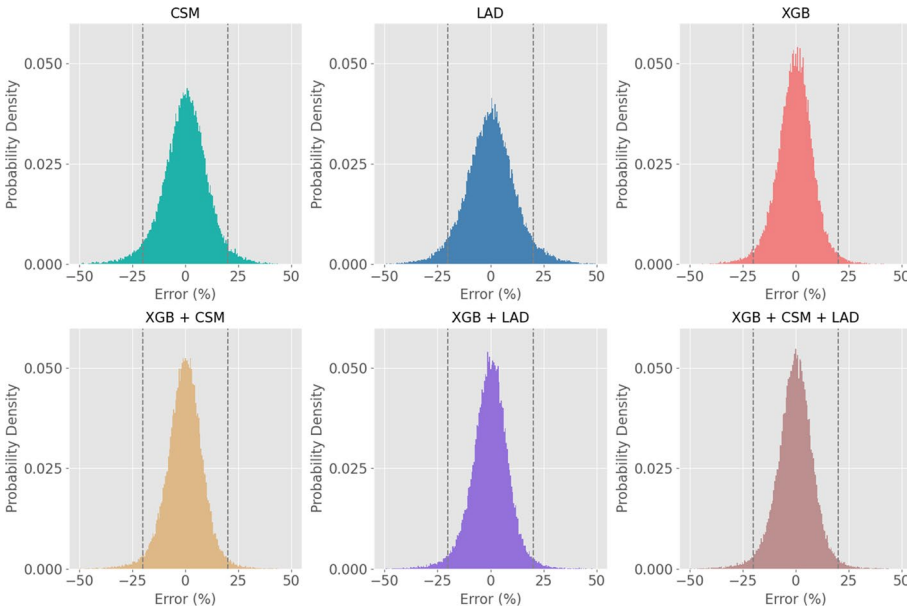

Fig. 3 Distribution of relative prediction errors Note: The dashed vertical lines mark the +/- 20% range

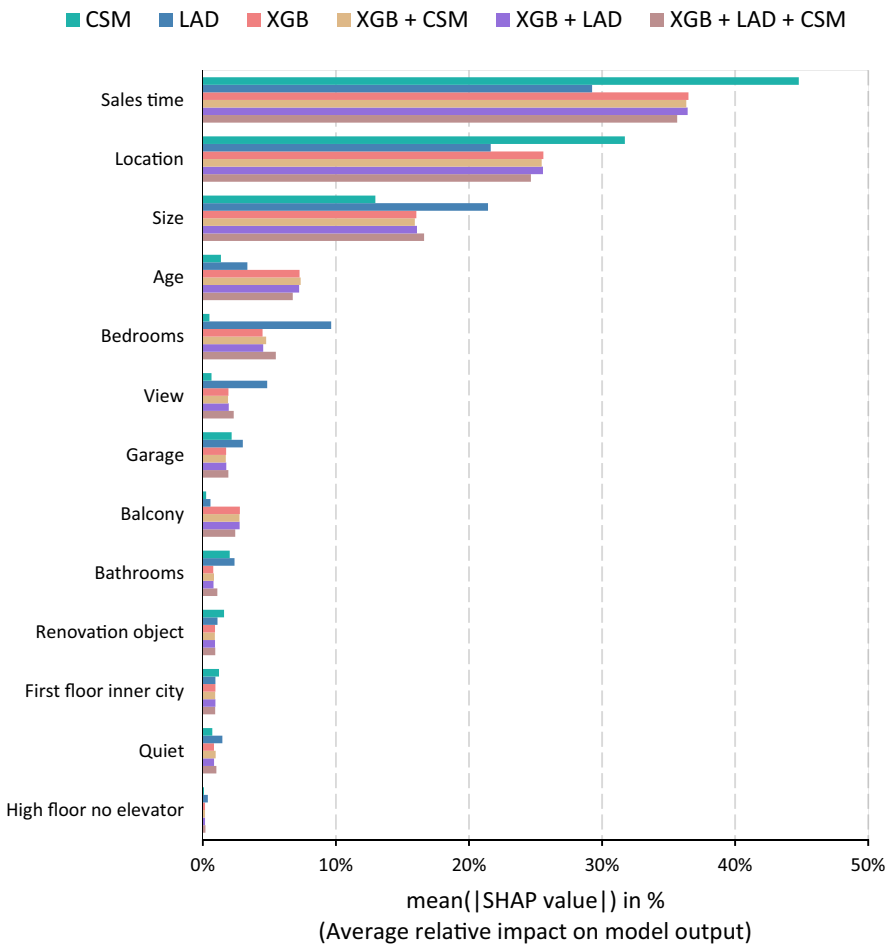

Fig. 4 Mean absolute SHAP values in percent from all models. Note: This figure shows the mean absolute relative SHAP contribution of various variables across different models. For the stacked models, which use predictions from base models as inputs, the contributions of individual variables are determined by decomposing the SHAP values of the base models and weighting them according to each base model's influence within the stacked ensemble. The time feature incorporates all categorical variables pertaining to year and quarter fixed effects for the LAD model, whereas CSM and XGB employ the *Days elapsed* variable. The location feature encompasses all categorical district variables in the LAD model, while XGB employs latitude and longitude, and CSM employs latitude and longitude represented in radians. All other variables are common for all models

The *Sales time* feature is the most influential variable across the models, followed by *Location* and *Size*. However, for the CSM, *Location* and *Sales time* switch ranks. Notably, the CSM and LAD models diverge significantly from the others in terms of variable weighting. While the top three variables remain consistent with the other models, LAD assigns lower importance to *Sales time* and *Location*, and

higher importance to *Size*. Conversely, CSM emphasizes *Sales time* and *Location* more strongly, while assigning less importance to *Size*. Although overall feature importances are lower, similar deviations are observed for variables such as *Age*, *Bedrooms*, *View*, *Balcony*, and *Bathrooms*. The variable *High floor and no elevator* is consistently ranked as the least important across all models.

Beeswarm plots from each model are found in Fig. 7 in the Appendix. The stacked models heavily rely on the predictions from their base models, with XGB predictions consistently dominating the SHAP value ranges, suggesting that XGB provides the strongest signal in the ensemble. Meanwhile, the contribution of LAD in stacked models is relatively small, indicating that its inclusion offers limited added value.

Model Performance at District Level

Our investigation also involved the identification of any potential patterns across districts. Figure 5 shows the average absolute error rate in each district, with a darker shade indicating a smaller prediction error. Our findings suggest that the predictive models perform better in the eastern region of Oslo compared to the western region.

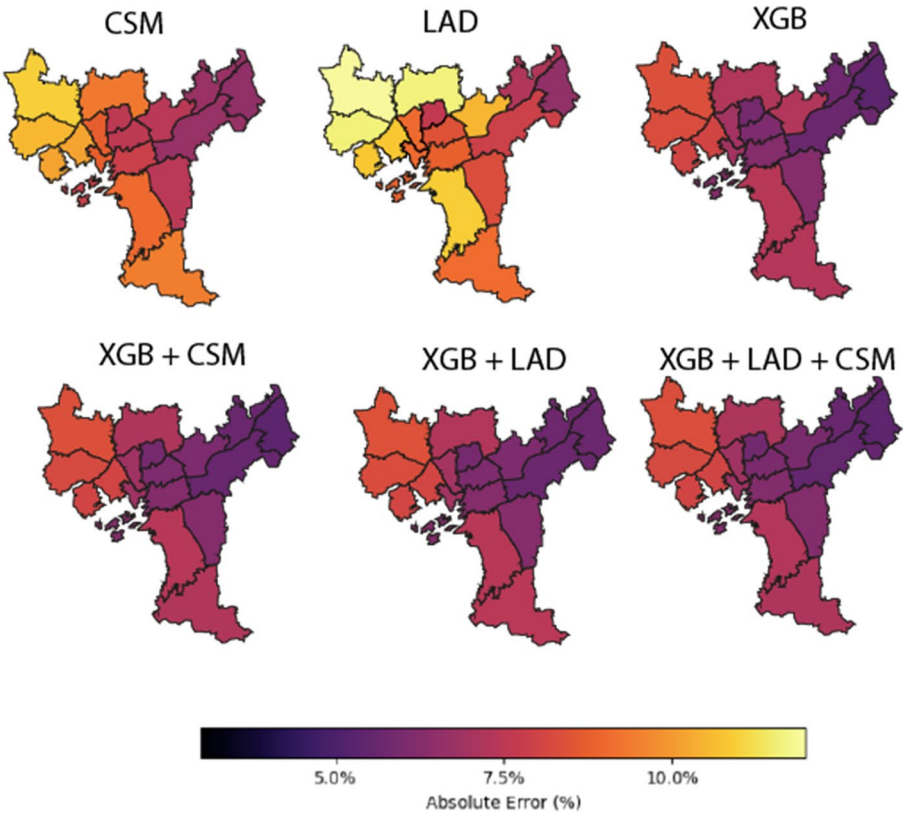

Fig. 5 Absolute percentage error per district from all models

One possible explanation for this result is that apartments in the eastern region are more homogeneous, mostly consisting of apartment complexes with similar sizes and prices, while the western region possesses more variation due to the generally higher income levels. Additionally, our results reveal that micro-location is a critical factor when predicting prices, which may explain why the LAD model performed poorly. Notably, the prediction errors observed in the XGB model and the three stacked models are very similar, further substantiating the effectiveness of the XGB model.

Model Performance with Different Sample Sizes

To gain a deeper understanding of how the different valuation techniques as well as the stacking perform under different data sizes, we conduct additional tests to simulate such scenarios. Our objective is to evaluate the models' ability to generalize well to new data with fewer observations and determine their adaptability to limited or scarce data. A model's adaptability to smaller datasets reflects its robustness and flexibility, which are essential qualities given that real-world scenarios often require dealing with data constraints. The nature of the test is essentially a stress test for the models, evaluating their capabilities to adapt to less-than-ideal circumstances.

Previous research, such as Stang et al. (2022), has shown that traditional methods may outperform the XGB model in areas with limited observations, suggesting that the XGB model may be more sensitive to information constraints compared to the CSM and the LAD model. Thus, we aim to investigate how our models perform out-of-sample on a variety of smaller training sets, which represent a wide range of potential scenarios. To achieve this, we begin with the original training set of 123,464 observations and systematically halve its size ten times, reducing it incrementally to as few as 121 observations ($\sim$0.1% of the original training sample). To minimize the influence of randomness, we maintain a constant test set size of 41,155 observations throughout the experiment. Figure 6 illustrates how the MdAPE differentiates across the various sizes of training data.

All models show improvement in MdAPE as the training set grows, but their rates of improvement vary. LAD starts with the worst performance at small training sizes and improves rapidly but plateaus after around 15,000 observations. CSM improves steadily but remains the least accurate model until about 30,000 observations, where it overtakes LAD. In contrast, XGB consistently outperforms both LAD and CSM, achieving the lowest MdAPE among the base models. Notably, CSM and XGB continue improving at a similar rate with larger training sets, albeit with diminishing returns.

The stacked models consistently outperform the individual models, illustrating the benefit of combining predictions. Among them, XGB + CSM performs best for smaller training sets (up to about 7,000 observations). However, with larger training sets, the fully stacked model slightly outperforms or matches the performance of XGB + CSM. By contrast, XGB + LAD performs worse than the other stacked models and is often comparable to the individual XGB. Interestingly, the inclusion

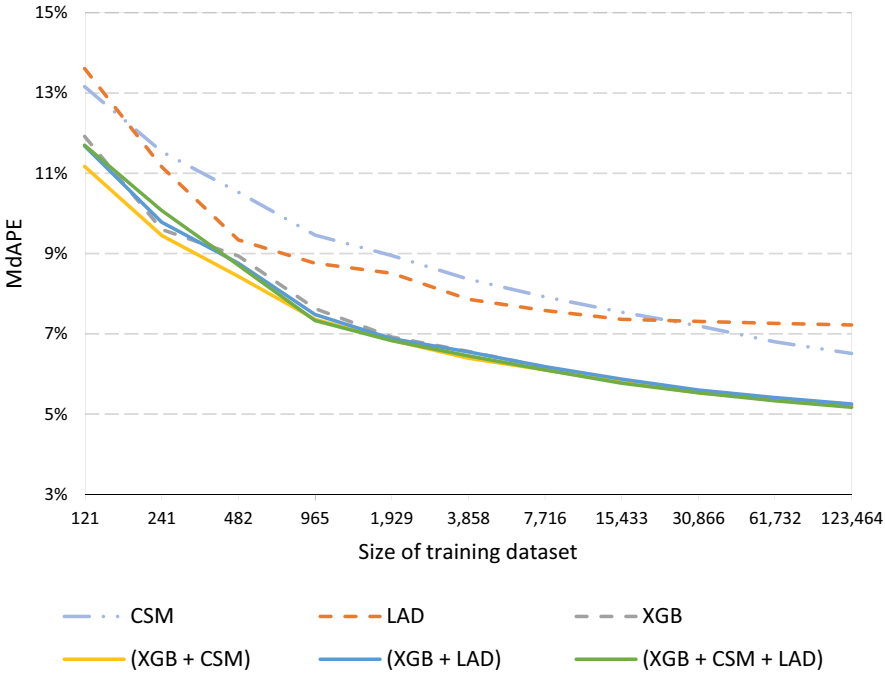

Fig. 6 MdAPE for training samples of varying sizes. Note: The training samples range from 121 observations (~0.1%) to the original training sample of 123,464 observations (100%). For training sets with fewer than 1,000 observations, interactions in the LAD model are excluded to mitigate the challenges posed by high dimensionality in a limited training sample. The horizontal axis is logarithmic (base 2)

of LAD in stacked models appears to contribute negatively for larger training sets, as evidenced by the slightly worse performance of the fully stacked model compared to XGB + CSM. Although stacked models perform best overall, their advantage over the individual XGB diminishes after around 2,000 observations, suggesting that CSM and LAD provide little additional value in stacked generalization. Similar trends are observed in the other evaluation metrics, mirroring those seen with MdAPE.[4]

A possible reason for XGB's sustained precision could be its ability to leverage a greater amount of information about each individual property compared to LAD, specifically in terms of location and time of sale. While CSM also incorporates these factors better than LAD, it becomes vulnerable when the sample size is too small. This may be due to its reliance on having enough similar properties to the target property.

Model Performance in Out-of-Time Prediction

One potential drawback of our models is that they do not incorporate out-of-time predictions, which are necessary to evaluate their performance in real-world

[4] These results are available from the authors upon request.

situations. Since the models have varying reliance on time and considering that we generally want to use the most recent data possible in a valuation setting, we conduct an out-of-time prediction by selecting the 2,000 most recent observations as the test sample. Meanwhile, the training set consists of the first 162,619 observations. The results are presented in Table 4.

As expected, the overall performance drops somewhat when dealing with the time perspective. However, we observe that the models' relative performance remains mostly consistent to the models trained with the original approach, as shown in Table 3. XGB remains the best-performing individual model, while LAD continues to fall short. Furthermore, the two stacked models with CSM still outperform the individual XGB across all evaluation metrics, whereas the XGB + LAD model fails to offer any improvement over XGB. While the results are somewhat less distinct, our main conclusion remains consistent. A notable difference, however, is that stacking only XGB and CSM now yields slightly better results on MdAPE, PPE(20), and AVMU compared to stacking all three models. This suggests that LAD may contribute negatively to out-of-time predictions.

Conclusion

This study develops and compares six distinct Automated Valuation Models (AVMs) to assess their prediction accuracy in the housing market. By employing stacked generalization, which combines methodologically diverse models, we aim to enhance the overall performance of the ensemble. Furthermore, we analyze how each model within the stacked ensemble function to identify the most suitable conditions for their use.

Our findings confirm that XGBoost (XGB) stands out as the most effective individual model, consistently demonstrating robust results when tested on an extensive sample from Oslo's housing market. Notably, XGB achieves a low median absolute percentage error (MdAPE) of 5.24%. Significantly, when incorporating the Comparable Sales Method (CSM) and our linear model,

Table 4 Comparison of model performance in out-of-time prediction

Models	R^2	RMSE	MAPE	MdAPE	PPE(10)	PPE(20)	PPE(50)	AVMU
CSM	0.8301	0.1163	8.99	6.90	66.75%	91.05%	99.55%	10.98%
LAD	0.8008	0.1259	9.94	8.02	59.50%	88.90%	99.80%	11.68%
XGB	0.8988	0.0897	7.10	5.66	75.70%	95.80%	**99.90%**	**8.89%**
(XGB + CSM)	0.9011	0.0887	6.81	**5.26**	78.10%	**96.30%**	**99.90%**	**8.89%**
(XGB + LAD)	0.8961	0.0909	7.18	5.65	75.95%	95.65%	**99.90%**	8.96%
(XGB + CSM + LAD)	**0.9013**	**0.0886**	**6.79**	5.28	**78.20%**	96.25%	**99.90%**	8.96%

The test set comprises the 2,000 most recent observations. R^2 and RMSE are calculated using the log-transformed observed and predicted values, while MAPE, MdAPE, PPEs, and AVMU are calculated after reverting values to their original scale (Price per m^2). Highlighted numbers indicate best performance.

Least Absolute Deviation (LAD), through stacked generalization, we observe improvements across all evaluation metrics, with MdAPE decreasing to 5.17%. While the overall performance expectedly drops in out-of-sample predictions, the models' relative performance remains largely consistent, and our main conclusion holds.

When analyzing models trained on smaller samples, we find that the linear LAD model improves its MdAPE performance up to a training sample size of approximately 15,000 observations, after which it plateaus. XGB achieves most of its prediction improvement by around 20,000 observations but continues to show gradual gains with larger training samples. As expected, the accuracy of the CSM heavily depends on having a substantial amount of data. CSM surpasses the performance of the linear LAD model once the training sample reaches approximately 30,000 observations. This outcome aligns with expectations, as the linear model struggles to fully utilize available information, particularly for location and sales date. With smaller training samples, CSM performs poorly due to its reliance on data from properties that significantly differ from the one being valued. As the sample size grows, CSM, like XGB, is able to base valuations on more similar properties, thereby better leveraging the available information for each observation.

Overall, this study highlights the effectiveness of non-linear machine learning AVMs; while stacked generalization does improve prediction accuracy by combining diverse models, the XGB model performs exceptionally well on its own, raising the question of whether stacked generalization is worth the added complexity and resources required for training and prediction. Nevertheless, this approach builds a bridge between conventional appraisal practices, such as the Comparable Sales Method employed by appraisers in the field, and traditional linear valuation models like hedonic models, as well as non-linear valuation techniques driven by machine learning. It also underscores the strengths and limitations of each individual model, which are all firmly rooted in the foundational concept of sales comparison, providing insights into the optimal conditions for their use. These findings have practical implications for real estate professionals, investors, and homebuyers, contributing to a deeper understanding of valuation techniques in the housing market.

Appendix

Table 5 LAD in-sample Estimation results

Variables	Coefficients
Size (ln)	−0.5627***
	(0.0017)
Age (ln)	−0.0291***
	(0.0004)
Bedrooms	0.1108***
	(0.0007)
Bathrooms	0.0531***
	(0.0008)
Balcony	−0.0023**
	(0.0009)
Garage	0.0416***
	(0.0009)
Quiet	−0.0256***
	(0.0008)
View	0.0817***
	(0.0008)
Renovation object	−0.1395***
	(0.0015)
High floor no elevator	0.0484***
	(0.0017)
First floor inner city	−0.0504***
	(0.0014)
Year × Quarter FE	Yes
District FE	Yes
Year × Housing attributes	Yes
District × Housing attributes	Yes
Observations	123,464
R^2	0.8780

The coefficients represent the effect on the natural logarithm of price per m^2 from the LAD model. The model has fixed effects for the interaction between year and quarter, as well as city district fixed effects. The model also includes the interaction between year and the remaining housing attributes, and the interaction between district and the other attributes. This allows for price dynamics to vary across time and space. Standard errors in parentheses. *Significant at the 1% level. **Significant at the 5% level.

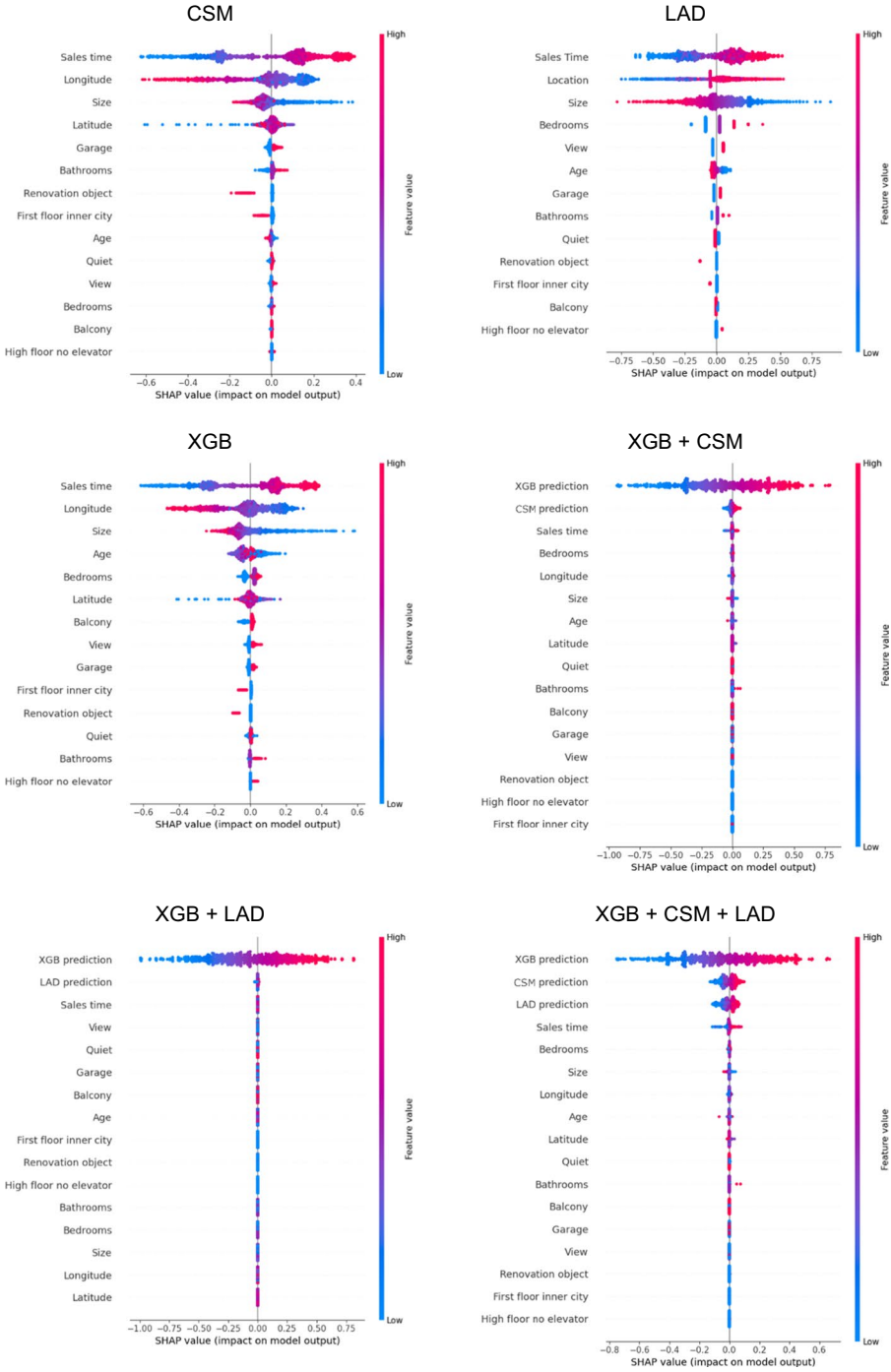

Fig. 7 SHAP beeswarm plots for all models

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Declarations

Ethical Approval Not applicable.

Competing Interests There are no competing interests.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

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